Ioannis Mitliagkas, associate professor **Curriculum vitae, November 2026**

CONTACT INFORMATION

Department of Computer Science and Operations Research
University of Montréal

Mila, Quebec AI Institute
6666 St. Urbain, Montréal

E-mail: ioannis@iro.umontreal.ca
Web: mitliagkas.github.io

RESEARCH INTERESTS

Statistical machine learning, optimization, deep learning, game theory, generative models, distributed learning systems.

AFFILIATIONS

University of Montréal, Department of Computer Science and Operations Research

Associate professor — June 2022 -
Assistant professor — Sept. 2017 - May 2022

Mila, Quebec AI Institute — September 2017 -

Core member, Canada CIFAR AI chair
Member of Mila's Scientific Council

Google DeepMind, Montréal, Staff Research Scientist (part time) — September 2022-

Archimedes, AI Institute — January 2024-

Affiliate researcher

PAST AFFILIATIONS

ElementAI, Montréal, Faculty Fellow (part time) — 2018-2020

Stanford University, Postdoc, Departments of CS and Statistics — 2015-2017
Supervised by: Associate prof. Christopher Ré, Adjunct prof. Lester Mackey

EDUCATION

The University of Texas at Austin

PhD, ECE department. — Awarded in August 2015
Advised by: Prof. Constantine Caramanis and Prof. Sriram Vishwanath
Thesis topic: Resource-Constrained, Scalable Learning

Technical University of Crete, Chania, Greece

MSc., ECE department. — 2008 - 2010

Diploma, Electronic and Computer Engineering (5 year degree), — 2002 - 2008
Advisor: Professor Nikos D. Sidiropoulos

Research grants

During my first seven years of tenure-track work, I was awarded a total of about 90 PhD years in competitive funding (about 2.8 million CAD after subtracting any overhead), to be disbursed over a period of 10 years.

- Google x Mila Research grant, awarded 2025-09 — $60,000
- Meta collaborative grant, awarded 2024-03 — $91,000
- CIFAR Canada AI chair, (started April 2018): Phase II awarded 2023-04, until 2028 — $1,050,000
- Google x Mila Research grant, awarded 2023-09 — $50,000
- Microsoft Research collaborative grant, awarded 2021-01 — $54,000
- Samsung-Mila collaborative three-year grant, awarded 2020-09 — $240,000 received (of $2,400,000 total consortium funding)
- IVADO Postdoctoral Scholarship, for my postdoc K. Ahuja, awarded 2020-12 — $150,000
- CIFAR Catalyst Grant, in collaboration with Manuela Girotti and Murat Erdogdu (UofT, Vector), awarded 2020-09 — $0 received (co-applicant; $50,000 total)
- IVADO Postdoctoral Scholarship, Fellow tier, for my postdoc N. Loizou, awarded 2020-01 — $180,000
- Microsoft Research collaborative grant, awarded 2019-08 — $54,000
- NSERC Discovery, awarded 2019-04 (+ competitive accelerator supplement) — $127,500
- CIFAR Canada AI chair, awarded 2018-04 — $1,052,500
- Fonds de Recherche du Québec, Nature et technologies, Nouveau Chercheur, awarded 2018-04 — $50,800
- IVADO startup grant (Apogée), awarded 2017-09 — $91,875
- Université de Montréal startup grant, awarded 2017-09 — $70,000

Program leadership

Professional MSc in Machine Learning, Université de Montréal and Mila 2018 -

Professor responsible for the program; co-responsible with Guillaume Rabusseau since 2019.

Every student completes an industry internship, most of them partly funded through Mitacs Accelerate. As program lead, I have been the named principal investigator on over 80 of these internships since 2020. They are program grants, not research funding, and are not counted under Research grants.

Internships by sector, with example projects:

- Energy and utilities: solar-irradiance and load forecasting, hydro-turbine and alternator monitoring, power-grid planning, smart buildings
- Health and life sciences: retinal-image diagnostics, brain-lesion detection in MRI, burn assessment for first responders, assistive vision for blind people
- Machine learning methods: AutoML and hyperparameter tuning, efficient inference, differential privacy, causal discovery

- Media and entertainment: automatic trailer generation, haptic effects from film, game exploration with reinforcement learning, generative fashion imagery
- Drug discovery and chemistry: molecular representation and search, drug–target interaction, SARS-CoV-2 therapeutics
- Finance and insurance: volatility surfaces, high-frequency forecasting, insurance claims from images, real-estate valuation
- Recommender systems: federated recommendation on mobile devices, restaurant and food-ordering recommendation
- Vision and robotics: inspection robots, semi-supervised and active-learning object detection
- Industrial processes: acoustic fault detection, bioprocess forecasting, electronics quality control
- Language and speech: dialogue systems, factuality of LLM outputs, document information extraction, speech to intent
- Other: 5G network monitoring, autonomous driving and road conditions, groundwater prediction, crop-weed identification, supply-chain traceability, video tutoring

Doctoral students

Ryan D'Orazio (PhD candidate)

Divyat Mahajan (PhD candidate)

Zichu Liu (PhD candidate)

Ahmed (Mehdi) Inane (PhD candidate)

Mehrab Hamidi (PhD candidate)

Yukti Makhija (PhD student)

Mahdi Ghaznavi (PhD student, starting January '27)

Past students and supervisees

Nicolas Loizou (postdoctoral scholar, 2021-12 → assistant prof. at Johns Hopkins Univ.)

Manuela Girotti (postdoctoral scholar, 2021-07 → assistant prof. at Emory University, Atlanta, GA)

Kartik Ahuja (postdoctoral scholar, 2022-09 → research scientist at FAIR Paris)

Kilian Fatras (postdoctoral scholar, 2024-01 → research scientist at Dreamfold, Montreal)

Alexia Jolicoeur-Martineau (PhD, 2023-01 → research scientist, Microsoft Research)

Reyhane Askari Hemmat (PhD, 2024-05 → Meta, Montreal)

Adam Ibrahim (PhD, 2024-05 → Google DeepMind, Mountain View)

Charles Guille-Escuret (PhD, 2025-01 → MBZUAI Institute of Foundation Models)

Hiroki Naganuma (PhD, 2026-07 → NVIDIA)

Vincent Quirion (MSc, 2024-2026 → Soma Energy)

Mehrnaz Mofakhami (MSc, 2024-08 → research assistant at Mila)

Rémi Piché-Taillefer (MSc, 2021-06 → Microsoft Research, Montréal)

Brady Neal (MSc, 2019-12 → Senior Research Scientist at Dataiku)

Oscar Clivio (research intern, 2024-11 → PhD student, University of Oxford)

Lyle Kim (research intern, 2023-01 → PhD candidate, Rice University)

Vincent Quirion (research intern, 2024-08 → MSc at UdeM/Mila, in progress)

Ange-Clement Akazan (research intern, 2023-12 → MSc student at the African Institute of Mathematical Sciences)

Séb Arnold (research intern, 2018-08 → PhD candidate at USC)

Vinayak Tantia (research intern, 2018-08 → research engineer, FAIR Montréal)

Baptiste Goujaud (research intern, 2020-03 → PhD candidate, École Polytechnique)

Amartya Mitra (research intern, 2020-06 → ML Researcher at Morgan Stanley)

Nicolas Gagné (research intern, 2018-08 → PhD candidate at McGill)

Teaching

University of Montreal Latest teaching evaluation: 3.7/4.0

Theoretical principles for deep learning	**Winter 2027**
Fundamentals of machine learning (in french)	**Fall 2026**
Fundamentals of machine learning (in french)	**Fall 2025**
Fundamentals of machine learning (in french)	**Fall 2024**
Fundamentals of machine learning	**Fall 2023**
Theoretical principles for deep learning	**Winter 2023**
Fundamentals of machine learning	**Fall 2022**
Theoretical principles for deep learning	**Winter 2022**
Fundamentals of machine learning	**Fall 2021**
Theoretical principles for deep learning	**Winter 2021**
Fundamentals of machine learning	**Fall 2020**
Theoretical principles for deep learning	**Winter 2020**
Fundamentals of machine learning	**Fall 2019**
Theoretical principles for deep learning	**Winter 2019**
Fundamentals of machine learning	**Fall 2018**
Theoretical principles for deep learning	**Winter 2018**

The University of Texas at Austin

Teaching Assistant—Information Theory	**Spring 2012**

Technical University of Crete

Teaching Assistant—Telecommunication Networks	**Fall 2008**

Publications

Z. Liu, N. Okolo, R. D'Orazio, D. Vucetic, **I. Mitliagkas**, G. Gidel
A Surrogate Perspective on Convergence of Fixed-Target DQN
Neural Information Processing Systems (NeurIPS), 2026

A.M. Inane, V. Quirion, G.K. Dziugaite, **I. Mitliagkas**
Unlearning with Asymmetric Sources: Improved Unlearning-Utility Trade-off with Public Data
arXiv:2605.11170
International Conference on Machine Learning (ICML), 2026

H. Naganuma, S. Gupta, Y. Briki, **I. Mitliagkas**, I. Rish, P. Raman, HJ.M. Shi
Adaptive Batch Sizes Using Non-Euclidean Gradient Noise Scales for Stochastic Sign and Spectral Descent
arXiv:2602.03001
International Conference on Machine Learning (ICML), 2026

S. Nakai, Y. Fadhloun, K. Mathlouthi, K. Yoshida, G. Talluri, **I. Mitliagkas**, H. Naganuma
Generalization Measures under Controlled Covariate Shift: A Regime-Aware Benchmark
arXiv:2602.01718
Transactions on Machine Learning Research (TMLR), 2026

D. Mahajan, S. Goyal, B. Youbi Idrissi, M. Pezeshki, **I. Mitliagkas**, D. Lopez-Paz, K. Ahuja
Beyond Multi-Token Prediction: Pretraining LLMs with Future Summaries
arXiv:2510.14751
International Conference on Learning Representations (ICLR), 2026

N. Tsikouras, Y. Pantis, **I. Mitliagkas**, C. Tzamos
A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond
arXiv:2510.19382
International Conference on Learning Representations (ICLR), 2026

K. Yoshida, Y. Naraki, T. Horie, R. Shimizu, **I. Mitliagkas**, H. Naganuma
DisTaC: Conditioning Task Vectors via Distillation for Robust Model Merging
International Conference on Learning Representations (ICLR), 2026

H. Naganuma, R. Hataya, K. Yoshida, **I. Mitliagkas**
An Empirical Study of Pre-trained Model Selection for Out-of-Distribution Generalization and Calibration
arXiv:2307.08187
Transactions on Machine Learning Research (TMLR), 2025

TH. Zhang, L. Maes, A. Milligan, A. Jolicoeur-Martineau, **I. Mitliagkas**, D. Scieur, S. Lacoste-Julien, C. Guille-Escuret
Understanding Adam Requires Better Rotation Dependent Assumptions
arXiv:2410.19964
Neural Information Processing Systems (NeurIPS), 2025

H. Naganuma, X. Zhang, MC. Yue, **I. Mitliagkas**, PA. Witte, RJ. Hewett, YT. Lee
Pseudo-Asynchronous Local SGD: Robust and Efficient Data-Parallel Training
arXiv:2504.18454
Transactions on Machine Learning Research (TMLR), 2025

D. Mahajan, M. Pezeshki, C. Arnal, **I. Mitliagkas**, K. Ahuja, P. Vincent
Compositional risk minimization
arXiv:2410.06303
International Conference on Machine Learning (ICML), 2025

R. D'Orazio, D. Vucetic, Z. Liu, JL. Kim, **I. Mitliagkas**, G. Gidel
Solving hidden monotone variational inequalities with surrogate losses
arXiv:2411.05228
International Conference on Learning Representations (ICLR), 2025

D. Beaglehole, **I. Mitliagkas**, A. Agarwala
Feature learning as alignment: a structural property of gradient descent in non-linear neural networks
arXiv:2402.05271
Transactions on Machine Learning Research (TMLR), 2024

E. Triantafillou, P. Kairouz, F. Pedregosa, J. Hayes, M. Kurmanji, K. Zhao, V. Dumoulin, J.J. Junior, **I. Mitliagkas**, J. Wan, L.S. Hosoya, S. Escalera, G.K. Dziugaite, P. Triantafillou, I. Guyon
Are we making progress in unlearning? Findings from the first NeurIPS unlearning competition
arXiv:2406.09073
Neural Information Processing Systems (NeurIPS), 2024

C. Guille-Escuret, PA. Noël, **I. Mitliagkas**, D. Vazquez, J. Monteiro
Expecting The Unexpected: Towards Broad Out-Of-Distribution Detection
arXiv:2308.11480
Neural Information Processing Systems (NeurIPS), 2024

C. Guille-Escuret, H. Naganuma, K. Fatras, **I. Mitliagkas**
No Wrong Turns: The Simple Geometry Of Neural Networks Optimization Paths
arXiv:2306.11922
International Conference on Machine Learning (ICML), 2024

D. Mahajan, **I. Mitliagkas**, B. Neal, V. Syrgkanis
Empirical Analysis of Model Selection for Heterogeneous Causal Effect Estimation
arXiv:2211.01939
International Conference on Learning Representations (ICLR), 2024 [**spotlight presentation**]

S. Lachapelle, D. Mahajan, **I. Mitliagkas**, S. Lacoste-Julien
Additive Decoders for Latent Variables Identification and Cartesian-Product Extrapolation
arXiv:2307.02598
Neural Information Processing Systems (NeurIPS), 2023 [**oral presentation**]

C. Guille-Escuret, P. Rodriguez, D. Vazquez, **I. Mitliagkas**, J. Monteiro
CADet: Fully Self-Supervised Out-Of-Distribution Detection With Contrastive Learning
arXiv:2210.01742
Neural Information Processing Systems (NeurIPS), 2023

T. Salvador, K. Fatras, **I. Mitliagkas**, A. Oberman
A Reproducible and Realistic Evaluation of Partial Domain Adaptation Methods
arXiv:2210.01210
Transactions on Machine Learning Research (TMLR), 2023

R. D'Orazio, N. Loizou, I. Laradji, **I. Mitliagkas**
Stochastic Mirror Descent: Convergence Analysis and Adaptive Variants via the Mirror Stochastic Polyak Stepsize
arXiv:2110.15412
Transactions on Machine Learning Research (TMLR), 2023

S. Lachapelle, T. Deleu, D. Mahajan, **I. Mitliagkas**, Y. Bengio, S. Lacoste-Julien, Q. Bertrand
Synergies Between Disentanglement and Sparsity: Generalization and Identifiability in Multi-Task Learning
arXiv:2211.14666
International Conference on Machine Learning (ICML), 2023

H. Naganuma, K. Ahuja, S. Takagi, T. Motokawa, R. Yokota, K. Ishikawa, I. Sato, **I. Mitliagkas**
Empirical Study on Optimizer Selection for Out-of-Distribution Generalization
arXiv:2211.08583
Transactions on Machine Learning Research (TMLR), 2023

R. Askari Hemmat*, A. Mitra*, G. Lajoie, **I. Mitliagkas**
LEAD: Min-Max Optimization from a Physical Perspective
arXiv:2010.13846
Transactions on Machine Learning Research (TMLR), 2023 Featured Certification; invited presentation at ICLR 2024

S. Sokota, R. D'Orazio, J. Z. Kolter, N. Loizou, M. Lanctot, **I. Mitliagkas**, N. Brown, C. Kroer
A Unified Approach to Reinforcement Learning, Quantal Response Equilibria, and Two-Player Zero-Sum Games
arXiv:2206.05825
International Conference on Learning Representations (ICLR), 2023

A. Mousavi-Hosseini, S. Park, M. Girotti, **I. Mitliagkas**, M. A. Erdogdu
Neural Networks Efficiently Learn Low-Dimensional Representations with SGD
arXiv:2209.14863
International Conference on Learning Representations (ICLR), 2023

M. Mofakhami, **I. Mitliagkas**, G. Gidel
Performative Prediction with Neural Networks
arXiv:2304.06879
Artificial Intelligence and Statistics (AISTATS), 2023

C. Guille-Escuret, A. Ibrahim, B. Goujaud, **I. Mitliagkas**
Gradient Descent Is Optimal Under Lower Restricted Secant Inequality And Upper Error Bound
arXiv:2203.00342
Neural Information Processing Systems (NeurIPS), 2022

K. Fatras, H. Naganuma, **I. Mitliagkas**
Optimal transport meets noisy label robust loss and MixUp regularization for domain adaptation
arXiv:2206.11180
Conference on Lifelong Learning Agents (CoLLAs), 2022

K. Ahuja, D. Mahajan, V. Syrgkanis, **I. Mitliagkas**
Towards efficient representation identification in supervised learning
arXiv:2204.04606
Conference on Causal Learning and Reasoning (CLeaR), 2022

N. Loizou, H. Berard, G. Gidel, **I. Mitliagkas**, S. Lacoste-Julien
Stochastic Gradient Descent-Ascent and Consensus Optimization for Smooth Games: Convergence Analysis under Expected Co-coercivity
arXiv:2107.00052
Neural Information Processing Systems (NeurIPS), 2021

K. Ahuja, E. Caballero, D. Zhang, JC. Gagnon-Audet, Y. Bengio, **I. Mitliagkas**, I. Rish
Invariance Principle Meets Information Bottleneck for Out-of-Distribution Generalization
arXiv:2106.06607
Neural Information Processing Systems (NeurIPS), 2021 [**spotlight presentation**]

A. Jolicoeur-Martineau, R. Piche-Taillefer, **I. Mitliagkas**, R. Tachet des Combes
Adversarial score matching and improved sampling for image generation
arXiv:2009.05475
International Conference on Learning Representations (ICLR), 2021

C. Guille-Escuret*, B. Goujaud*, M. Girotti, **I. Mitliagkas**
A Study of Condition Numbers for First-Order Optimization
arXiv:2012.05782
Artificial Intelligence and Statistics (AISTATS), 2021 [best student paper award OPT2020]

G. K. Dziugaite, A. Drouin, B. Neal, N. Rajkumar, E. Caballero, L. Wang, **I. Mitliagkas**, D. Roy
In search of robust measures of generalization
arXiv:2010.11924
Neural Information Processing Systems (NeurIPS), 2020

N. Loizou, H. Berard, A. Jolicoeur-Martineau, P. Vincent, S. Lacoste-Julien, **I. Mitliagkas**
Stochastic Hamiltonian Gradient Methods for Smooth Games
arXiv:2007.04202
International Conference on Machine Learning (ICML), 2020

A. Ibrahim, W. Azizian, G. Gidel, **I. Mitliagkas**
Linear Lower Bounds and Conditioning of Differentiable Games
arXiv:1906.07300
International Conference on Machine Learning (ICML), 2020

W. Azizian, D. Scieur, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel
Accelerating Smooth Games by Manipulating Spectral Shapes
arXiv:2001.00602
Artificial Intelligence and Statistics (AISTATS), 2020

W. Azizian, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel
A Tight and Unified Analysis of Gradient-Based Methods for a Whole Spectrum of Differentiable Games
arXiv:1906.05945
Artificial Intelligence and Statistics (AISTATS), 2020

S. M. Arnold, P. A. Manzagol, R. Babanezhad, **I. Mitliagkas**, N. L. Roux
Reducing the variance in online optimization by transporting past gradients
arXiv:1906.03532
Neural Information Processing Systems (NeurIPS), 2019 [**spotlight presentation**]

I. Albuquerque, J. Monteiro, T. Doan, B. Considine, T. Falk, **I. Mitliagkas**
Multi-objective training of Generative Adversarial Networks with multiple discriminators
arXiv:1901.08680
International Conference on Machine Learning (ICML), 2019

V. Verma, A. Lamb, C. Beckham, A. Najafi, **I. Mitliagkas**, A. Courville, D. Lopez-Paz, Y. Bengio
Manifold Mixup: Better Representations by Interpolating Hidden States
arXiv:1806.05236
International Conference on Machine Learning (ICML), 2019

A. Lamb, J. Binas, A. Goyal, S. Subramanian, **I. Mitliagkas**, Y. Bengio, M. Mozer
State-Reification Networks: Improving Generalization by Modeling the Distribution of Hidden Representations
arXiv:1905.11382
International Conference on Machine Learning (ICML), 2019 [**oral presentation**]

B. Kanuparthi, D. Arpit, G. Kerg, N. R. Ke, **I. Mitliagkas**, Y. Bengio
h-detach: Modifying the LSTM Gradient Towards Better Optimization
arXiv:1810.03023
International Conference on Learning Representations (ICLR), 2019

G. Gidel*, R. Askari Hemmat*, M. Pezeshki, R. Le Priol, G. Huang, S. Lacoste-Julien, **I. Mitliagkas**
Negative Momentum for Improved Game Dynamics
arXiv:1807.04740
Artificial Intelligence and Statistics (AISTATS), 2019

J. Zhang, **I. Mitliagkas**
YellowFin and the Art of Momentum Tuning
arXiv:1706.03471
Systems and Machine Learning (SysML), 2019 [earlier version, titled YellowFin: Adaptive optimization for (A)synchronous systems, presented as an oral at SysML 2018] [**oral presentation**]

P. Achlioptas, O. Diamanti, **I. Mitliagkas**, L. Guibas
Learning Representations and Generative Models for 3D Point Clouds
arXiv:1707.02392
International Conference on Machine Learning (ICML), 2018

C. De Sa, B. He, **I. Mitliagkas**, C. Ré, P. Xu
Accelerated stochastic power iteration
arXiv:1707.02670
Artificial Intelligence and Statistics (AISTATS), 2018

J. Zhang, **I. Mitliagkas**
YellowFin: Adaptive optimization for (A)synchronous systems
Systems and Machine Learning (SysML), 2018 [**oral presentation**]

T. Kurth, J. Zhang, N. Satish, **I. Mitliagkas**, E. Racah, M.A. Patwary, T. Malas, N. Sundaram, W. Bhimji, M. Smorkalov, J. Deslippe, M. Shiryaev, S. Sridharan, P. Prabhat, P. Dubey
Deep Learning at 15PF: Supervised and Semi-Supervised Classification for Scientific Data
arXiv:1708.05256
Supercomputing (SC), 2017

I. Mitliagkas, L. Mackey
Improving Gibbs Sampler Scan Quality with DoGS
arXiv:1707.05807
International Conference on Machine Learning (ICML), 2017

B. He, C. De Sa, **I. Mitliagkas**, C. Ré
Scan Order in Gibbs Sampling: Models in Which it Matters and Bounds on How Much
arXiv:1606.03432
Neural Information Processing Systems (NeurIPS), 2016

I. Mitliagkas, C. Zhang, S. Hadjis, C. Ré
Asynchrony begets Momentum, with an Application to Deep Learning
arXiv:1605.09774
Allerton Conference on Communication, Control, and Computing (Allerton), 2016

I. Mitliagkas, M. Borokhovich, A. Dimakis, C. Caramanis
FrogWild! – Fast PageRank Approximations on Graph Engines
arXiv:1502.04281
International Conference on Very Large Data Bases (VLDB), 2015

D. Papailiopoulos, **I. Mitliagkas**, A. Dimakis, C. Caramanis
Finding Dense Subgraphs via Low-Rank Bilinear Optimization
International Conference on Machine Learning (ICML), 2014

I. Mitliagkas, C. Caramanis, P. Jain
Memory-limited Streaming PCA
arXiv:1307.0032
Neural Information Processing Systems (NeurIPS), 2013

I. Mitliagkas, A. Gopalan, C. Caramanis, S. Vishwanath
User Rankings from Comparisons: Learning Permutations in High Dimensions
Allerton Conference on Communication, Control, and Computing (Allerton), 2011

I. Mitliagkas, N. D. Sidiropoulos, A. Swami
Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks: Convex Approximation and Distributed Implementation
IEEE Transactions on Wireless Communications (TWC), 2011

I. Mitliagkas, S. Vishwanath
Strong Information-Theoretic Limits for Source/Model Recovery
Allerton Conference on Communication, Control, and Computing (Allerton), 2010 [also circulated as Is Capacity Defined for Source Recovery & Learning Problems?]

I. Mitliagkas, N. D. Sidiropoulos, A. Swami
Distributed Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks
International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2010

I. Mitliagkas, N. D. Sidiropoulos, A. Swami
Convex Approximation-based Joint Power and Admission Control for Cognitive Underlay Networks
International Wireless Communications and Mobile Computing Conference (IWCMC), 2008

Workshop papers

R. D'Orazio, S. Anand, **I. Mitliagkas**, G. Gidel
Surrogate Loss Gradient Descent in Variational Inequalities with Linear Approximation
NeurIPS 2026 Workshop on Optimization for Machine Learning (OPT)

L. Gomezjurado Gonzalez, T. Nakamori, G. Talluri, A. Tiwari, H. Kawashima, **I. Mitliagkas**, G. Rabusseau, H. Naganuma
Orth-Dion: Eliminating Geometric Mismatch in Distributed Low-Rank Spectral Optimization
ICML 2026 CoLoRAI Workshop, arXiv preprint arXiv:2605.16341

O. Clivio, D. Mahajan, P. Taslakian, S. Magliacane, **I. Mitliagkas**, V. Zantedeschi, A. Drouin
Learning to Defer for Causal Discovery with Imperfect Experts
ICLR 2025 Workshop on Reasoning and Planning for LLMs, arXiv preprint arXiv:2502.13132

H. Naganuma, JL. Kim, A. Kyrillidis, **I. Mitliagkas**
Smoothness-Adaptive Sharpness-Aware Minimization for Finding Flatter Minima
ICLR 2024 Workshop on Practical ML for Limited/Low Resource Settings (PML4LRS)

M. Mofakhami, R. Bayat, **I. Mitliagkas**, J. Monteiro, V. Zantedeschi
Performance Control in Early Exiting to Deploy Large Models at the Same Cost of Smaller Ones
ICML 2024 workshop on Efficient Systems for Foundational Models, arXiv preprint arXiv:2412.19325

A. Ibrahim, C. Guille-Escuret, **I. Mitliagkas**, I. Rish, D. Krueger, P. Bashivan
Towards Out-of-Distribution Adversarial Robustness
ICML 2023 Workshop on New Frontiers in Adversarial Machine Learning (AdvML-Frontiers), arXiv preprint arXiv:2210.03150

B. Neal, **I. Mitliagkas**
In Support of Over-Parametrization in Deep Reinforcement Learning: an Empirical Study
ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena

B. Neal, S. Mittal, A. Baratin, V. Tantia, M. Scicluna, S. Lacoste-Julien, **I. Mitliagkas**
A Modern Take on the Bias-Variance Tradeoff in Neural Networks
ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena, arXiv preprint arXiv:1810.08591

J. Zhang, C. De Sa, **I. Mitliagkas**, C. Ré
Parallel SGD: When does averaging help?
Optimization Methods for the Next Generation of Machine Learning Workshop, ICML 2016, arXiv preprint arXiv:1606.07365

Preprints under review, technical reports

A.M. Inane, M. Haghifam, G.K. Dziugaite, **I. Mitliagkas**
Towards Faithful Data Selection: Statistical Limits and Scalable Algorithms
arXiv preprint arXiv:2610.30442

H. Naganuma, L. Gomezjurado Gonzalez, M. Ghaznavi, G. Talluri, A. Nitanda, S.P. Liew, R. Hataya, **I. Mitliagkas**
Which Geometry on Which Layer? Measured Sign-Spectral Assignment in Mixed-Optimizer Training
arXiv preprint arXiv:2610.17285

S. Dufort-Labbé, M. Hamidi, R. Pascanu, **I. Mitliagkas**, D. Scieur, A. Baratin
Navigating Potholes with Geometry-Aware Sharpness Minimization
arXiv preprint arXiv:2605.16134

AC. Akazan, A. Jolicoeur-Martineau, **I. Mitliagkas**
Generating Tabular Data Using Heterogeneous Sequential Feature Forest Flow Matching
arXiv preprint arXiv:2410.15516

M. Girotti, **I. Mitliagkas**, G. Gidel
Convergence Analysis and Implicit Regularization of Feedback Alignment for Deep Linear Networks
arXiv preprint arXiv:2110.10815

A. Jolicoeur-Martineau, K. Li*, R. Piché-Taillefer*, T. Kachman*, **I. Mitliagkas**
Gotta Go Fast When Generating Data with Score-Based Models
arXiv preprint arXiv:2105.14080

I. Albuquerque, J. Monteiro, T. Falk, **I. Mitliagkas**
Generalizing to unseen domains via distribution matching
arXiv preprint arXiv:1911.00804

A. Jolicoeur-Martineau, **I. Mitliagkas**
Gradient penalty from a maximum margin perspective
arXiv preprint arXiv:1910.06922

A. Ratner, D. Alistarh, G. Alonso, D. G. Andersen, P. Bailis, S. Bird, N. Carlini, B. Catanzaro, J. Chayes, E. Chung, B. Dally, J. Dean, I. S. Dhillon, A. Dimakis, P. Dubey, C. Elkan, G. Fursin, G. R. Ganger, L. Getoor, P. B. Gibbons, G. A. Gibson, J. E. Gonzalez, J. Gottschlich, S. Han, K. Hazelwood, F. Huang, M. Jaggi, K. Jamieson, M. I. Jordan, G. Joshi, R. Khalaf, J. Knight, J. Konečný, T. Kraska, A. Kumar, A. Kyrillidis, A. Lakshmiratan, J. Li, S. Madden, H. B. McMahan, E. Meijer, **I. Mitliagkas**, R. Monga, D. Murray, K. Olukotun, D. Papailiopoulos, G. Pekhimenko, T. Rekatsinas, A. Rostamizadeh, C. Ré, C. De Sa, H. Sedghi, S. Sen, V. Smith, A. Smola, D. Song, E. Sparks, I. Stoica, V. Sze, M. Udell, J. Vanschoren, S. Venkataraman, R. Vinayak, M. Weimer, A. G. Wilson, E. Xing, M. Zaharia, C. Zhang, A. Talwalkar
MLSys: The New Frontier of Machine Learning Systems
White paper, arXiv:1904.03257

A. Lamb, J. Binas, A. Goyal, D. Serdyuk, S. Subramanian, **I. Mitliagkas**, Y. Bengio
Fortified Networks: Improving the Robustness of Deep Networks by Modeling the Manifold of Hidden Representations
arXiv preprint arXiv:1804.02485

S. Hadjis, C. Zhang, **I. Mitliagkas**, C. Ré
Omnivore: An Optimizer for Multi-device Deep Learning on CPUs and GPUs
Technical report, arXiv:1606.04487

In the Press

The Greek AI Ecosystem

Trudeau meets with newly appointed Canada CIFAR AI Chairs, **CIFAR News**

Sept chercheurs de l'UdeM titulaires de chaires en intelligence artificielle Canada-CIFAR, **UdeM Nouvelles**

NERSC Scales Scientific Deep Learning to 15 Petaflops, **HPC Wire**

De la Grèce à l'UdeM: l'étonnant parcours d'Ioannis Mitliagkas, **UdeM Nouvelles**

Awards, Distinctions

Recognized among top reviewers of ICML, 2026

NeurIPS Foundation, top 10% of reviewers, 2020

CIFAR Canada AI chair

NIPS Foundation, listed among best reviewers, 2018

Gerondelis Foundation Inc.: Graduate Scholarship, 2014

The University of Texas at Austin: Microelectronics and Computer Development (MCD) Fellowship, 2009-2011

Technical University of Crete: Undergraduate excellence award, 2008

State Scholarships Foundation (Greece): Undergraduate excellence award, 2005

Technical Chamber of Greece: Undergraduate excellence award, 2005

Professional Service

Archimedes summer school, Athens, Greece (2022)

Neuromatch Academy Deep Learning Track, World-wide (2021, 2022)

AI4Good Summer School, Canada (2021)

Co-organizer of INFORMS 2021 workshop:
"Accelerated methods in convex and non-convex optimzation"

Co-founder and organizer of MTL MLOpt, a Montreal-based group of researchers from many universities and industrial labs.

Member of the inaugural program committee of MLSys:
The committee's role was to decide the conference's focus and steer its future goals.

Organizer of NeurIPS 2018, 2019 workshop:
"Smooth Games Optimization and Machine Learning"

Reviewer of MITACS Accelerate grants

Served as head of the scientific committee in charge of evaluating IVADO grants (2018) and as member in subsequent years.

Host of oral session, NeurIPS 2021

Reviewer and AC for a number of journals and conferences including NeurIPS, ICML, COLT, AISTATS, ALT, AAAI, ICLR, JMLR (editorial board), IJCAI, SIGGRAPH, Transactions on Information Theory, ISIT, ICASSP, Transactions on Wireless Communications.

Recent Invited Talks (not including accepted paper presentations)

Neuromatch Academy Deep Learning Track, World-wide	Summer 2023
Archimedes summer school, Athens, Greece	July 2022
Simons semester in game theory, Berkeley, CA	January-March, 2022
SIAM conference in optimization OP21, Spokane, WA	July 2021
Neuromatch Academy Deep Learning Track, World-wide	Summer 2021
AI4Good Summer School, Canada	May 2021
Google Brain, Montreal, QC	November 2020
ITA, San Diego, CA	February 2020
INFORMS, Seattle, WA	October 2019
Microsoft Research workshop, Montréal, QC	October 2019
Theoretical Advances in Deep Learning, Workshop, Istanbul	July 2019
UT Austin, TX	March 2019
NVIDIA, Webinar	March 2019
ElementAI, Toronto, ON	November 2018
BorealisAI, Toronto, ON	October 2018
USC, Los Angeles, CA	October 2018
Microsoft Research workshop, Montréal, QC	October 2018
ElementAI, Toronto, ON	September 2018
Microsoft Research, Montréal, QC	August 2018
ElementAI, Montréal, QC	June 2018
FAIR, Montréal, QC	May 2018
RLLab, McGill, Montréal, QC	April 2018
ElementAI, Montréal, QC	April 2018
TechAide, Montréal, QC	April 2018
ECE Seminar, UT Austin, TX	March 2018
BayesComp, Barcelona Spain	March 2018
SysML, Stanford CA	February 2018
Google Brain, Montréal	November 2017
Texas Wireless Summit, Austin, TX	October 2017
Colloquium, University of Montréal	September 2017
Colloquium, The University of Texas, Austin	September 2017
AutoML workshop, ICML, Sydney	August 2017
Workshop on Advances in Computing Architectures, Stanford SystemX	April 2016
ITA workshop, San Diego, CA	February 2017
AAAI 2017 Workshop on Distributed Machine Learning	February 2017
Microsoft Research, Cambridge, UK	December 2016

SystemX Stanford Alliance Fall Conference	November 2016
Microsoft Research, New England	October 2016
Allerton Conference, Monticello, IL	September 2016
Google Brain, Mountain View, CA	August 2016
MIT Lincoln Labs, MA	August 2016
NVIDIA, Santa Clara, CA	July 2016